

Email Security Protocols

Module 19

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Hello friends I am Professor yogesh Chaba from Department of computer science and engineering Guru jambheshwar University of Science and Technology Hisar harana in this lecture we will study about email security protocols first of all please see the contents which I shall cover after covering brief introduction about email security I shall take up different email security protocols like pretty good privacy pgp privacy enhanced mail pem and secure multi-purpose internet mail extension s mine in last we will compare all these protocols electronic mail most commonly referred to as email since 1993 is a method of exchanging digital messages from one sender to one or more recipients email operates across the internet or across other computer

networks email privacy is the broad topic dealing with issues of unauthorized assess and inspection of electronic mail this unauthorized assess can happen while an email is in transit as well as when it is stored on email server or on a user computer email has to go through potentially untrusted intermediate servers before reaching its destination and there is no way to tell if it was assessed by an unauthorized user or not email system is different from a letter sealed in an envelope in which by close inspection of the envelope it might be possible to tell if someone opened it or not in that sense an email is much like a postcard whose contents are visible to everyone who handles it there are certain technical methods that

make unauthorized assess to email difficult if not impossible however since email messages frequently cross Nation boundaries and different countries have different rules and regulations so email privacy is a complicated issue this problem has stimulated several people and groups to apply the cryptographic principles to produce secure email now we will study few common widely used email security systems pretty good privacy pgp privacy enhanced mail pem and secure multi-purpose internet mail extensions s mine first of all let us see pretty good privacy pgp short for pretty good privacy is a public key encryption program originally written by Phil Zimmerman in 1991 over the past few years pgp has got thousands of adherent supporters all

over the globe and has become a def facto standard for encryption of email on the internet released in 1991 pgp is a complete email Security package that provides privacy authentication digital signatures and compression all in an easy to use form furthermore the complete package including all the source code is distributed free of charge via the Internet due to its quality zero price and easily availability on Unix Linux Windows and Mac operating system platforms it is widely used today pgp can be used to send messages confidentially for this purpose pgp combines symmetric key encryption and public key encryption the message is encrypted using a symmetric encryption algorithm which requires a symmetric key each symmetric key is used only

once and is also called a session key the message and its session key are sent to the receiver so that they know how to decrypt the message but to protect it during transmission it is encrypted with the receiver's public key only the private key belonging to the receiver can decrypt the session key pgp supports message authentication and integrity checking the letter is used to detect whether a message has been altered since it was completed and the former to determine whether it was actually sent by the person or entity claimed to be the sender because the content is encrypted any changes in the message will result in failure of the decryption with the appropriate key 7 minutes, 1 second the sender uses pgp to create a digital signature for the message with either

the RSA or DSA algorithm to do so pgp computes a hash also known as a message digest from the plain text and then creates the digital signature from that hash using the senders private key to see how pgp works let us consider the example shown in figure here Alys on the left hand side wants to send a signed plain text message P to Bob on right hand side of figure in a secure way both Alys and Bob have RSA type of private and public key d and e let us assume that each one knows others public key Alis starts out by sending message P by invoking the pgp program on her computer by first hashing her message P using md5 and then encrypts the resulting hash using her private RSA key B A this signed hash of p is

concatenated with original message P resulting P1 which is transmitted after compressing by zip program giving output p1. Z on the receiver side Bob receives p1. Z message which is given to international data and encryption algorithm idea module for decryption when Bob eventually gets the message he can decrypt the hash with Ellis public key e and verify that the hash is correct after decompressing it Bob separates the plain text from the encrypted hash and decrypts the hash using Alis public key if the plain text hash agrees with his own md5 computation he knows that P is the correct message and it came from ellys after verification of hash function Bob reverses the base 64 encoding and decrypts the idea key using his private RSA

key it is worth to note that RSA is only used in two places here to decrypt the 128bit md5 hash and to encrypt the 128bit idea key although RSA is slow it has to encrypt only 256 bits not a large volume of data thus pgp provides security compression and a digital signature and does

so in a much more efficient way pgp supports four RSA key lengths it is up to the user to select the one that is most appropriate the lengths are number one casual 384 bits it can be broken easily today number two commercial 512 bits it is breakable by three letter organization number three military 1024 bits it is not breakable by anyone on this Earth number four Ellen 2048 bits it is not breakable by anyone on other planets also since RSA is only used for two

small computations everyone should use Ellen strength Keys all the time the format of a classic pgp message is as shown in figure numerous other formats are also in use the message has three parts containing the idea key the signature and the message respectively the key part contains the key identifier and key itself the signature part contains a signature header followed by a time stamp next field is the identifier for the senders public key that can be used to decrypt the signature hash and this field is followed by type information that identifies the algorithms used and the encrypted md5 hash itself the message part also contains a header followed by the default file name to be used if the receiver writes the file to the disk next field is message

creation Time stamp and finally the message itself this was about working of pgp algorithm now let us see a product that is open pgp which is being used nowadays based on pgp algorithm open pgp is the most widely used Email encryption standard in the world today it is defined by the open pgp working group of the internet engineering task force ietf standard RFC 480 the open pgp Alliance is a growing group of companies and other organizations that are implementers of the open pgp proposed standard the alliance works to facilitate technical interoperability and marketing Synergy between open pgp implementations the open pgp standard was originally derived from pretty good privacy pgp protocol RFC 480 document

contains all the necessary information to develop interoperable applications based on the open pgp format it describes the format and methods needed to read check generate and write conforming encrypted messages keys and signatures in July 1997 pgp incorporation proposed to the ietf that there should be a standard called open pgp they gave the ietf permission to use the name open pgp to describe this new standard as well as any program that supported the standard the ietf accepted the proposal and started the open pgp working group open pgp is on the internet standards track and is under active development specification of pgp was RFC 1991 but the current specification in open pgp is RFC 480 in 2007 the successor of RFC

24 the standard was extended to support ChaCha20 Cipher by RFC 581 in 2009 and encryption based on elliptic curve cryptography as RFC 637 in 2012 open pgps encryption can ensure secure delivery of files and messages as well as provide verification of who created or sent the message using a process called digital signing using open pgp for communication requires participation by both the sender and the recipient open pgp can also be used to secure sensitive files when they are stored in vulnerable places like mobile devices or in the cloud so after pgp let us move on to privacy enhanced mail pem in contrast to pgp which was initially a oneman show our second email security protocol privacy enhanced mail p is an official

internet standard and

described in four rfc's RFC 1421 through rfc4 424 it is a 1993 ietf proposal for securing email using public key cryptography very roughly PM covers the same territory as pgp that is privacy and authentication for RFC 822 based email systems nevertheless it also has some differences from pgp in approach and Technology PM is the outgrowth of work by the privacy and security research group psrg of the internet research task force irtf messages sent using p are first converted to a canonical form so that they all have the same conventions about white space like tabs and spaces next a message hash is computed using md2 or md5 then the concatenation of the hash and the message is encrypted using d s

in light of the known weakness of a 56-bit key this choice is certainly suspect the encrypted message can then be encoded with base 64 coding and transmitted to the recipient as in pgp each message is encrypted with a onetime key that is enclosed along with the message the key can be protected either with RSA or with tri triple Dees using Ed in pem Key Management is more structured as compared to pgp keys are certified by x.509 certificates issued by CA which are arranged in a rigid hierarchy starting at a single route the advantage of this scheme is that certific ific at revocation is possible by having the roote issue crls periodically the only problem with p is that nobody ever used it and it has gone

to that big bit bin in the sky although PM became an ietf proposed standard still it was never widely deployed or used the problem was largely political who would operate the route and under what conditions there was no shortage of candidates but many people were afraid to trust anyone company with the security of the whole system the most serious candidate RSA security incorporation wanted to charge per certificate issued however some organizations balked at this idea in particular the US government is allowed to use all us patents for free and companies outside the US had become accustomed to use the RSA algorithm for free as the company forget to patent it outside the the US no one was ready about suddenly having to pay RSS security

incorporation for doing something that they had always done for free in the end no route could be found and PM collapsed in addition to being an obstacle to deployment the single rooted hierarchy was rejected Ed by some commentators as an unacceptable imposition of central Authority this led Phil Zimmerman to propose the web of trust as the pki infrastructure for the encryption program pretty good privacy pgp efforts to deploy PM were finally abandoned in response to the need to to extend the protocol to support mime this led to the development of the protocol mime object Security Services moss and S mime which shares defao standard status with pgp this is an ietf standard a result of a group working for a long time the basic idea is to have privacy by virtue

of hierarchical authentication a receiver trusts the message of the sender when it is accompanied by a certificate from his trusted Authority these authoritative certificates are distributed from a group called internet policy registration Authority IP and policy certificate Authority PCA these trusted Authority actually certifies the public key sent by senders now after PM let us see third email security algorithm that is secure multi-purpose internet mail extensions s mine ietf next venture into email security known as secure multi-purpose internet mail extension s mine is described in RFC 2632 through 2643 as we know internet email messages consist of two parts the header and the body the header forms a collection of field or value pairs structured to

provide information essential for the transmission of the message the structure of these headers can be found in RFC 822 the body is normally unstructured unless the email is in mind format mime defines how the body of an email message is structured the MIM format permits email to include enhance Ed text Graphics audio and more in a standardized manner via MIM compliant mail systems however mime itself does not provide any Security Services the purpose of s mime is to Define such services like PM it provides authentication data Integrity secrecy and known reputation it also is quite flexible supporting a variety of cryptographic algorithms not surprisingly given the name s mime integrates with mime all

kind of messages to be protected secure multi-purpose internet mail extensions is a protocol that adds digital signatures and an encryption to internet mime messages a variety of new mime headers are defined for example for holding digital signatures ietf definitely learn something from the PM experience s mime does not have a rigid certificate hierarchy beginning at a single root instead user can have multiple trust anchors as long as certificate can be traced back to some trust anchor the user believes in it is considered valid SM mime uses the standard algorithms and protocols now in last let us see comparison of email security protocols we have studied three email security protocols first was pgp that is pretty

good privacy second was privacy enhanced mail pem and third was s mime as already discussed P could not gain popularity due to its shortcomings so it got collapsed pgp and S MIM which was adopted as an internet standard are two official email security systems which are currently National Institute of Standards and Technology nist specified standards both are considered to be secure mail transmission techniques there has been a great deal of confusion in public and even within the IET F about the status of S M and pgp let us compare s MIM version 3 and open pgp protocol which are widely used nowadays please see the figure although they offer similar services to the users the two protocols have many differences s mine supports binary

format whereas as open pgp supports pkcs7 mime format further and more important to corporate users they have different formats for their certificates as you can see s mime supports binary certificate based on x.509 version 3 whereas open pgp supports previous pgp based binary certificates this means that user of one protocol cannot communicate with the user

of other and also they cannot share authentication certificates s mime supports Dey Helman with DSS or RSA but open pgp supports elgl with DSS signature algorithm both both the protocol supports S1 hash algorithm s mime supports CMS format of mime encapsulation of signed data but open pgp supports asy the difference between the two protocols is similar to the differences

between giif and JPEG file they both do basically the same thing for end users but their formats are very different so friends this was about protocols for email Security in this lecture after covering brief introduction about email security we discussed different email security protocols like pretty good privacy pgp privacy enhanced mail p and secure multi-purpose internet mail extension s mine in last we also compared to most commonly used protocols s MIM version 3 and open pgp thank you